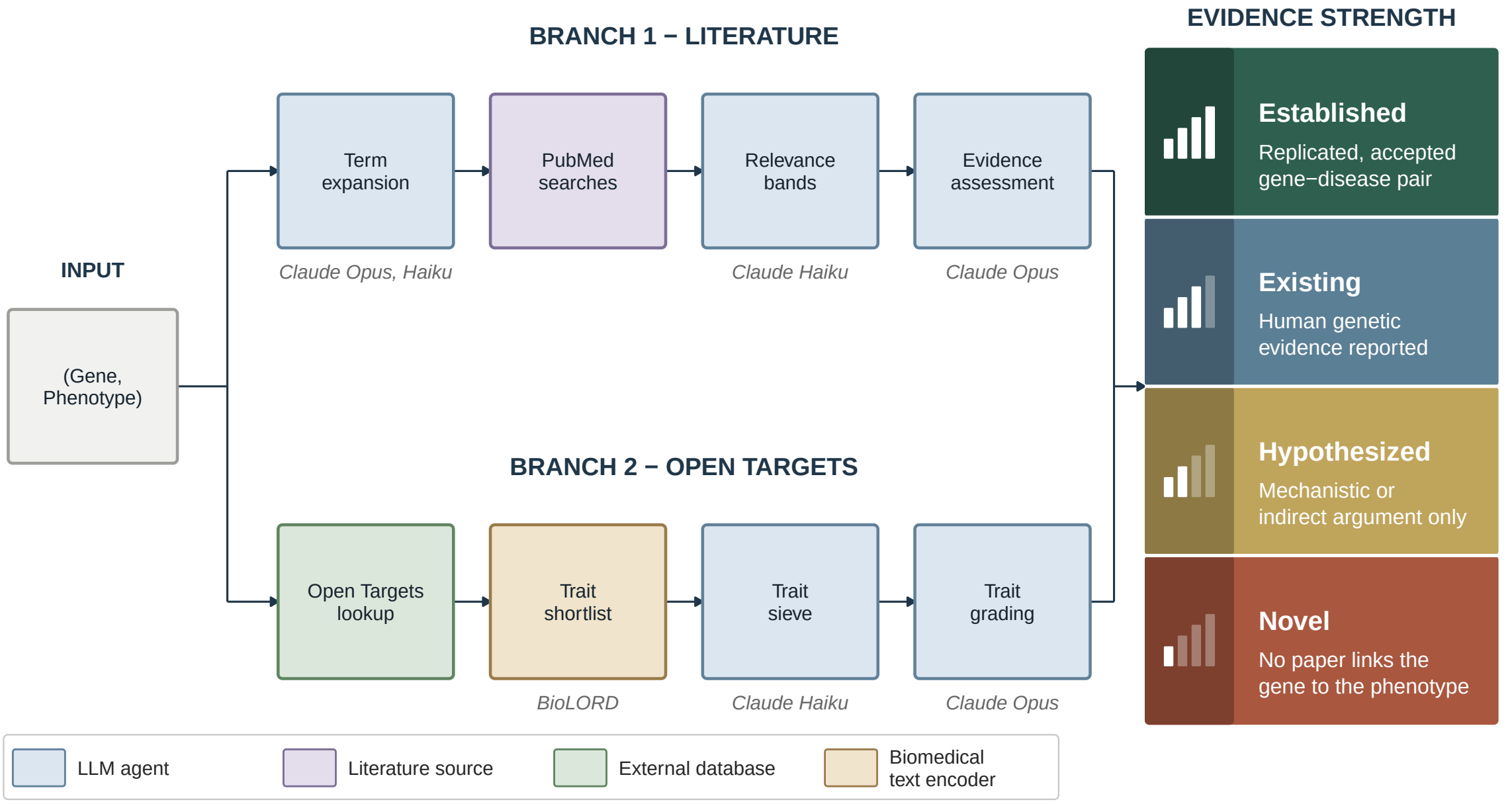
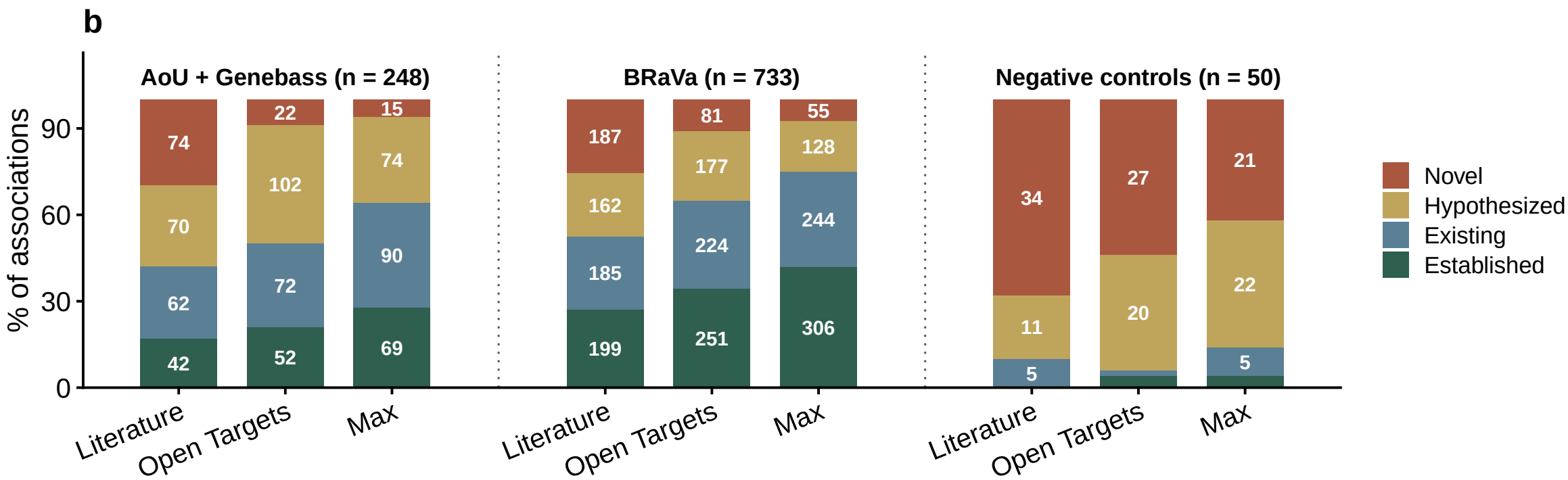
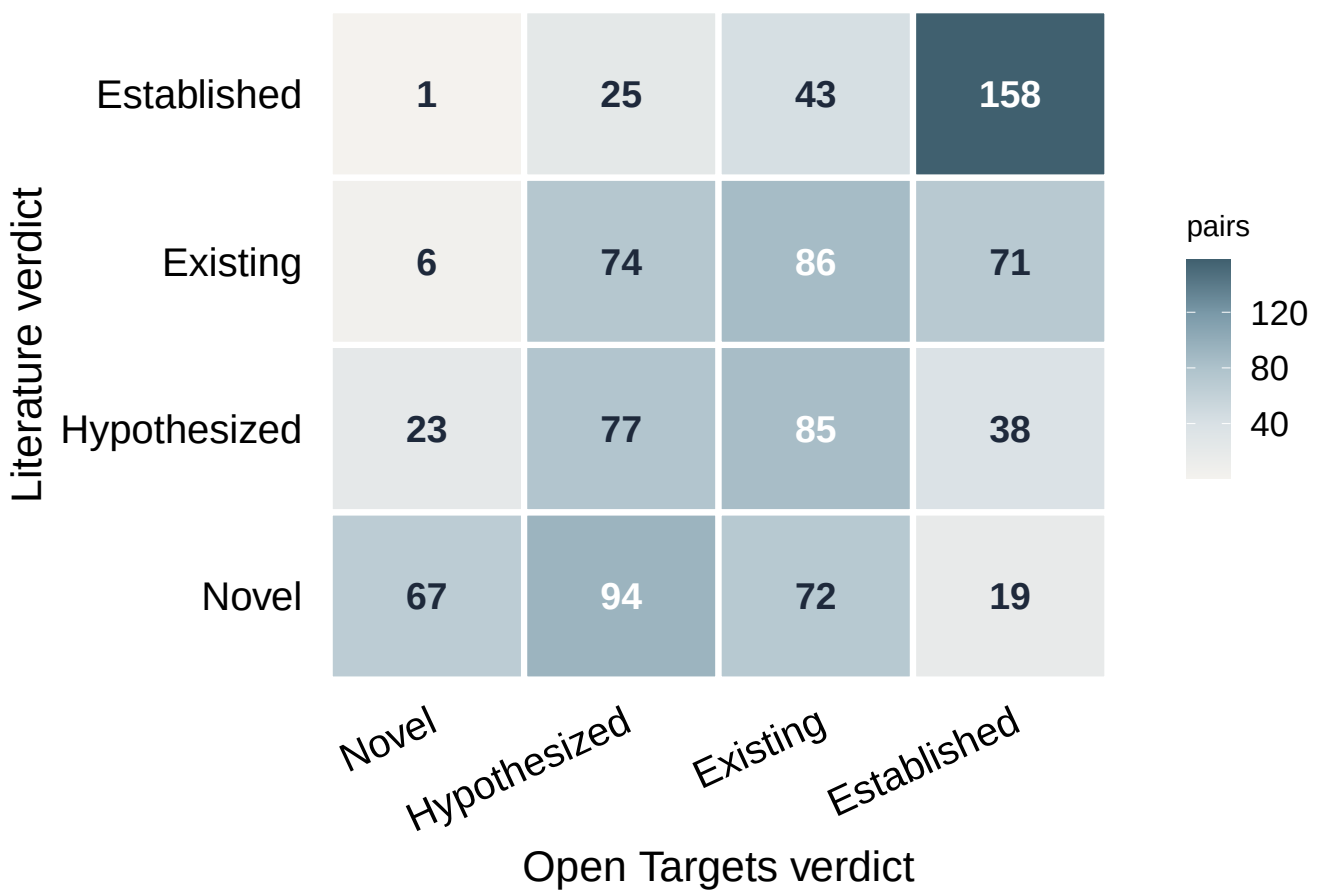
a


BRANCH 1 − LITERATURE
INPUT
(Gene, Phenotype)
Term expansion
PubMed searches
Relevance bands
Evidence assessment
Claude Opus, Haiku
Claude Haiku
Claude Opus
BRANCH 2 − OPEN TARGETS
Open Targets lookup
Trait shortlist
Trait sieve
Trait grading
BioLORD
Claude Haiku
Claude Opus
EVIDENCE STRENGTH
Established
Replicated, accepted gene−disease pair
Existing
Human genetic evidence reported
Hypothesized
Mechanistic or indirect argument only
Novel
No paper links the gene to the phenotype
LLM agent
Literature source
External database
Biomedical text encoder

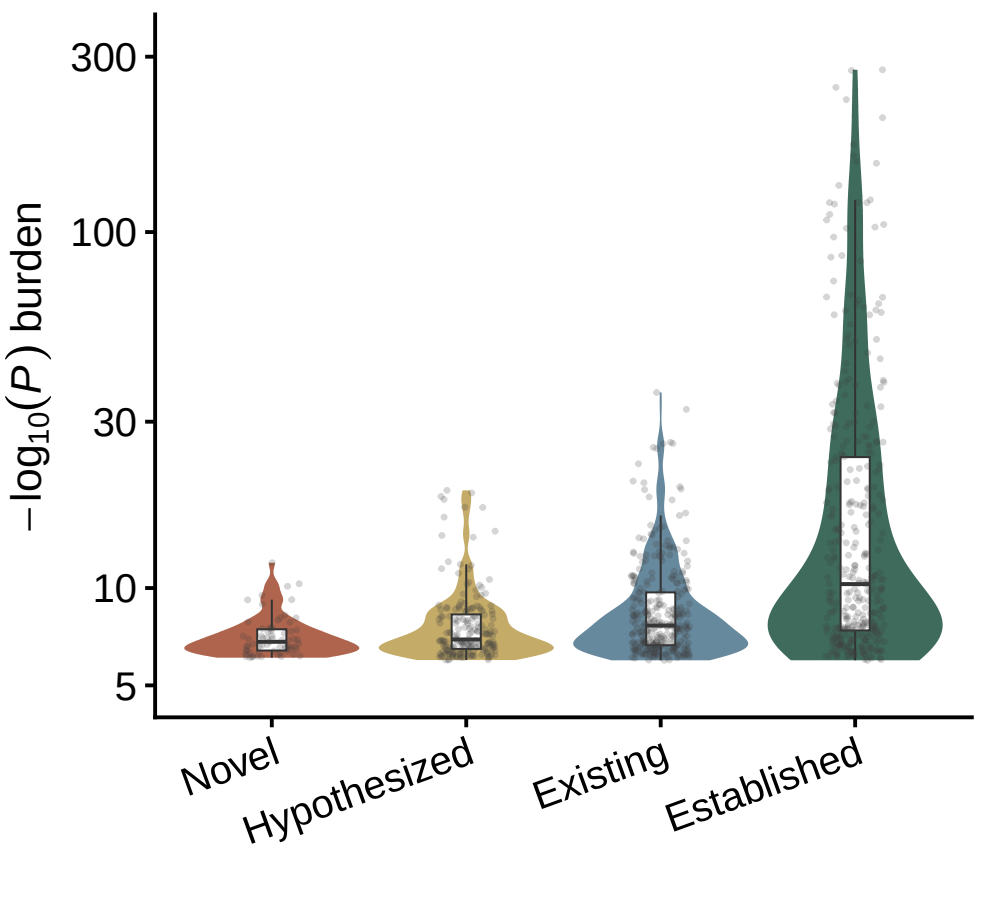
b


AoU + Genebass (n = 248)
BRaVa (n = 733)
Negative controls (n = 50)
% of associations
0
30
60
90
74
70
62
42
22
102
72
52
15
74
90
69
187
162
185
199
81
177
224
251
55
128
244
306
34
11
5
27
20
21
22
5
Literature
Open Targets
Max
Novel
Hypothesized
Existing
Established

c


Literature verdict
Established
Existing
Hypothesized
Novel
1
25
43
158
6
74
86
71
23
77
85
38
67
94
72
19
pairs
120
80
40
Novel
Hypothesized
Existing
Established
Open Targets verdict

d


−log10(P) burden
300
100
30
10
5
Novel
Hypothesized
Existing
Established